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1. Introduction to NoSQL Databases

What is NoSQL?

NoSQL (Not Only SQL) databases are databases that do not use traditional relational tables like MySQL or Oracle.

Instead they store data in:

- Documents
- Key-value pairs
- Graphs
- Columns

NoSQL databases are designed for:

- Big Data
- High performance
- Flexible schema
- Cloud applications

Types of NoSQL Databases

Type	Example	Data Format
Document Database	MongoDB	JSON
Key Value	Redis	key:value
Column Based	Cassandra	Columns
Graph	Neo4j	Nodes

SQL vs NoSQL

SQL	NoSQL

Table based	Document based
Fixed schema	Dynamic schema
Vertical scaling	Horizontal scaling
Example: MySQL	Example: MongoDB

Example

Student record.

SQL Database

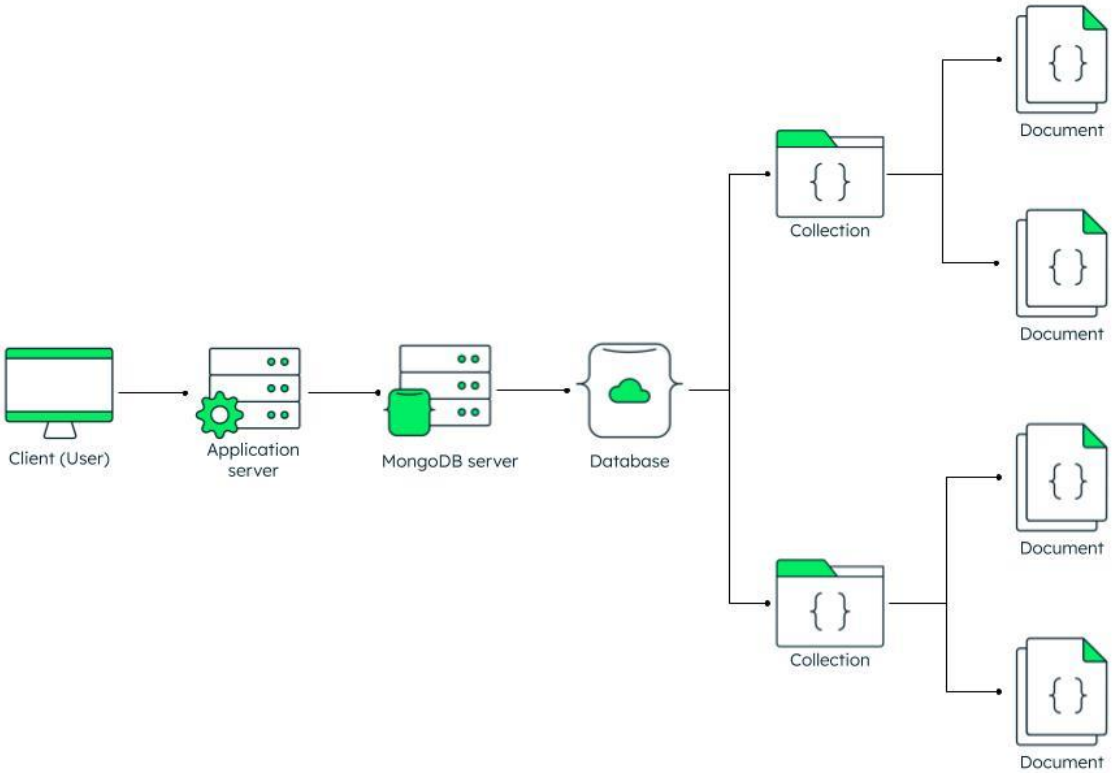
Id	**Name**	**course**
1	Rahul	CSE

MongoDB:

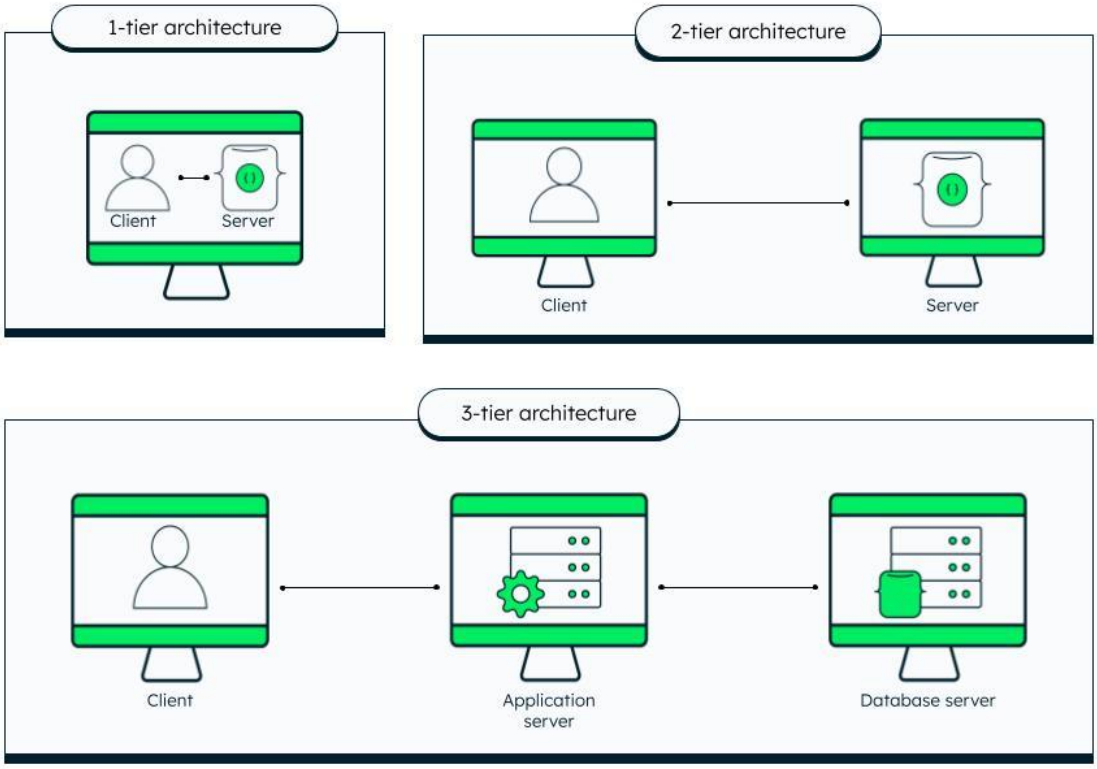
```
{
 "id":1,
 "name":"Rahul",
 "course":"BCA"
}
```

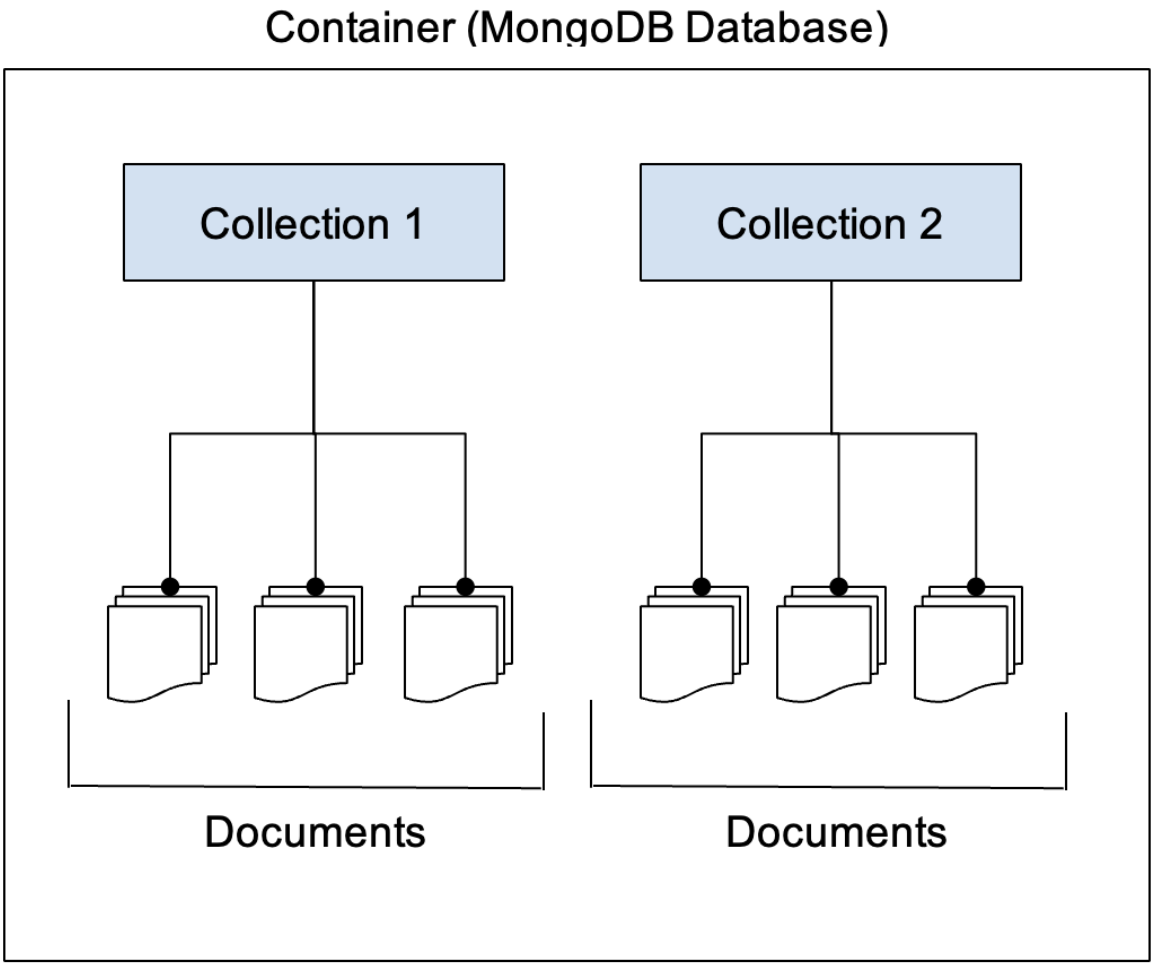
MongoDB Architecture

MongoDB basic architecture



Client server architecture





MongoDB Structure

Database

↓

Collection

↓

Document

↓

Fields

Example

CollegeDB

↓

Students

↓

```
{
 name:"Rahul",
 age:20
}
```

2. Installation and Configuration of MongoDB

Step 1 Download MongoDB

Download from:

https://www.mongodb.com/try/download/community

Step 2 Install MongoDB

Install using default settings.

MongoDB will install:

- MongoDB Server
- MongoDB Compass (GUI tool)

Step 3 Start MongoDB Server

Command:

Mongod

Step 4 Start Mongo Shell

Open another terminal:

Mongo

- Now MongoDB shell is ready.

Check Databases

show dbs

Example Output:

admin

local

config

3. CRUD Operations in MongoDB

CRUD means:

Operation	Meaning
C	Create
R	Read
U	Update
D	Delete

Create Database

MongoDB creates database automatically.

- use collegeDB

Create Collection

- db.createCollection("students")

Check collections

- show collections

Insert Data (CREATE)

- **Insert One Document**

```
db.students.insertOne({
 name:"Rahul",
 age:20,
 course:"BCA"
})
```

- **Insert Multiple Documents**

```
db.students.insertMany([
 {name:"Amit", age:22, course:"BBA"},
 {name:"Riya", age:21, course:"MCA"}
])
```

Read Data (SELECT)

- Show All Data

  ```
  db.students.find()
  ```

- Pretty Format

  ```
  db.students.find().pretty()
  ```

- Find Specific Data

  ```
  db.students.find({name:"Rahul"})
  ```

Update Data

Update One Document:

```
 db.students.updateOne(
 {name:"Rahul"},
 {$set:{age:23}}
)
```

Update Many Documents:

```
db.students.updateMany(
 {course:"BCA"},
 {$set:{course:"BCA Updated"}}
)
```

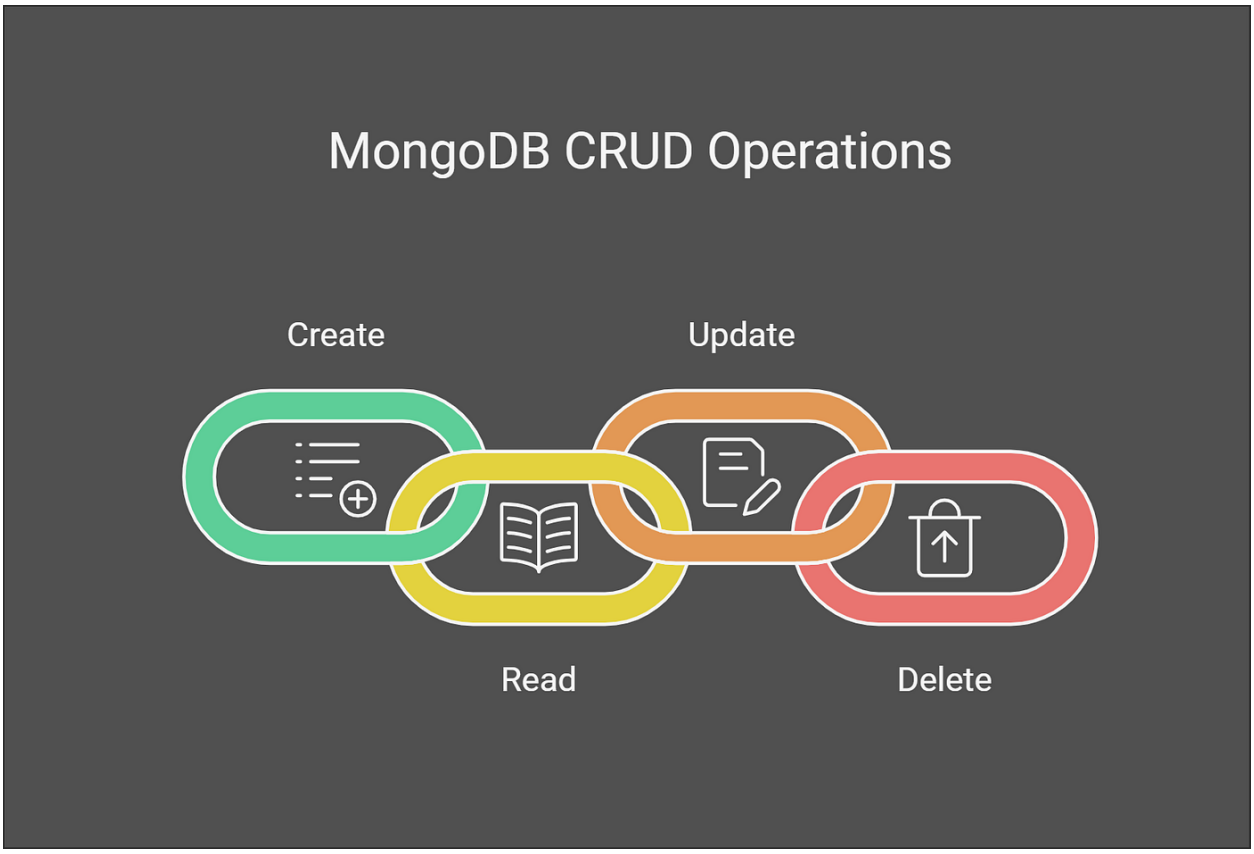
Delete Data:

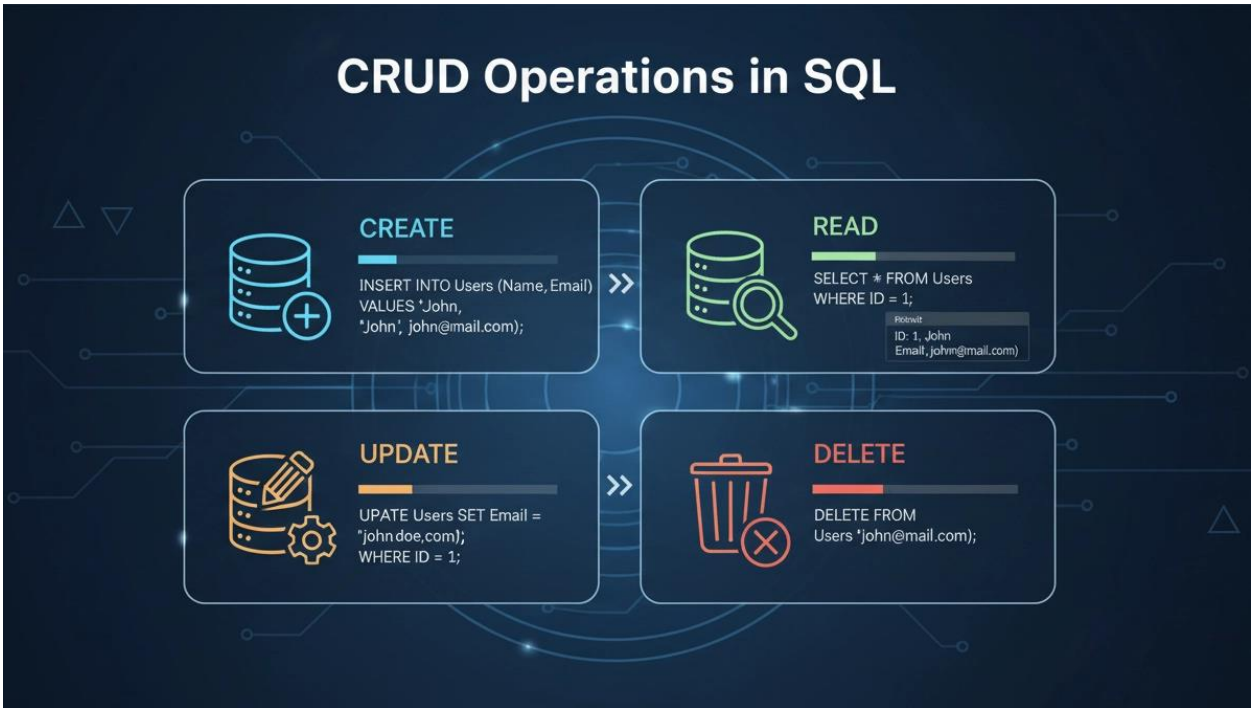
```
db.students.deleteOne({name:"Rahul"})
```

Delete Many Documents

```
db.students.deleteMany({course:"BBA"})
```

CRUD Flow Diagram





4. Indexing in MongoDB

What is Indexing?

Indexing improves database search speed.

Without index:

- Database checks every record.

With index:

- Database directly finds data.

Example Without Index

Search student name.

- 10000 records

Database checks each record, Slow.

Create Index

```
db.students.createIndex({name:1})
```

- 1 means **ascending order**

Descending Index

```
db.students.createIndex({age:-1})
```

Check Indexes

```
db.students.getIndexes()
```

5. Querying in MongoDB

Query means **searching data from database**.

Find Specific Field

```
db.students.find(
 {},
 {name:1, course:1}
)
```

Greater Than Query

```
db.students.find({age:{$gt:20}})
```

Less Than Query

```
db.students.find({age:{$lt:25}})
```

AND Query

```
db.students.find({
```

```
 age:{$gt:20},
 course:"BCA"
})
```

OR Query

```
db.students.find({
 $or:[
  {course:"BCA"},
  {course:"MCA"}
 ]
})
```

Query Operator Table

Operator	Meaning
$gt	Greater than
$lt	Less than
$gte	Greater than equal
$lte	Less than equal
$or	OR condition

6. Schema Design and Data Modeling

What is Schema?

Schema means structure of database data.

In MongoDB schema is flexible.

Example:

```
{
 name:"Rahul",
```

```
 age:21
}
```

Another document can have extra fields.

```
{
 name:"Amit",
 age:22,
 city:"Delhi"
}
```

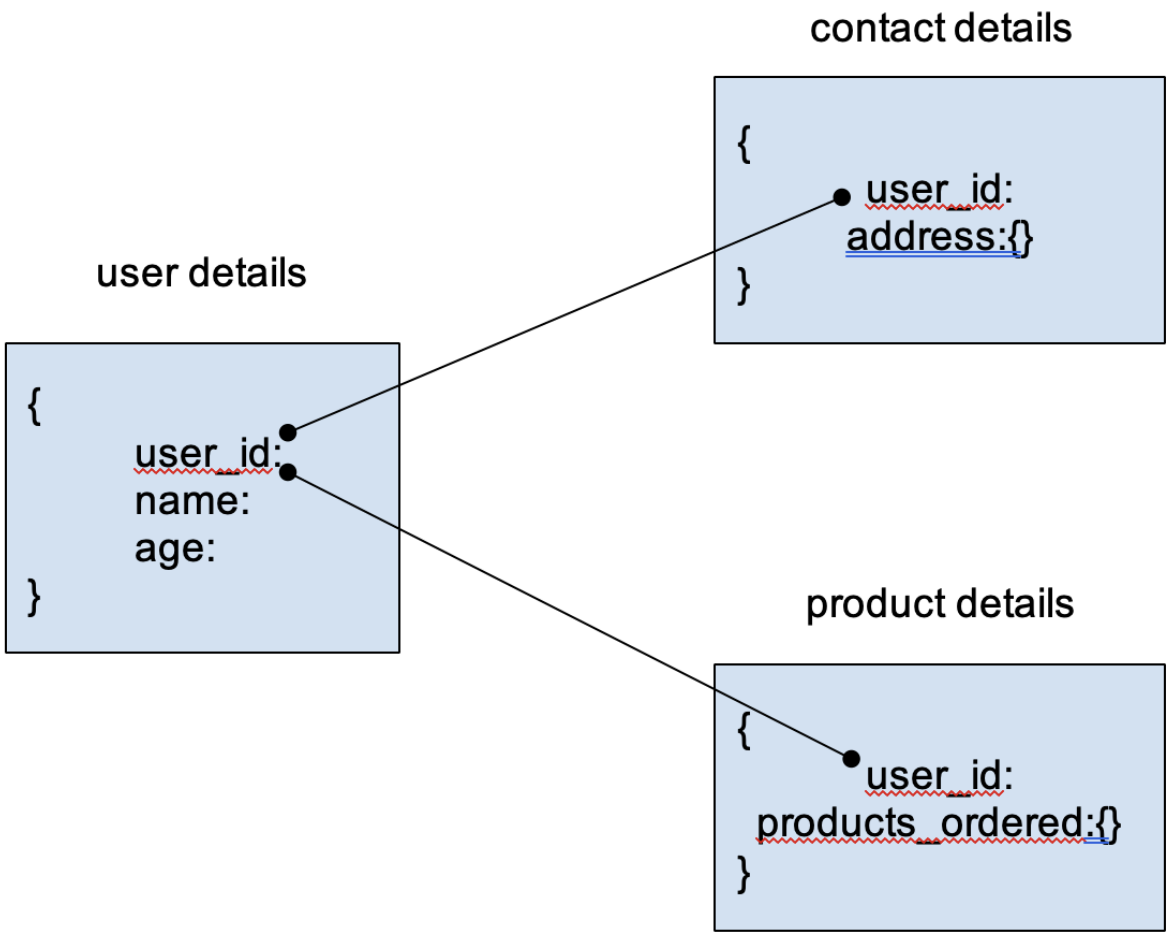
Data Modeling

Designing how data will be stored**.**

Two types:

1. Embedded Documents
2. Referenced Documents

Embedded Model



Example:

```
{
 name:"Rahul",
 course:"BCA",
 address:{
   city:"Lucknow",
   state:"UP"
 }
}
```

All data stored inside one document.

Advantages:

- Fast
- Simple queries

Reference Model

Example:

Students collection

```
{
 student_id:1,
 name:"Rahul"
}
```

Course collection

```
{
 course_id:101,
 course_name:"BCA",
 student_id:1
}
```

Here documents reference each other.

<u>Example Node.js + MongoDB Code:</u>

Install MongoDB driver

```
npm install mongodb
```

Example code:

```
const {MongoClient} = require("mongodb")

const url = "mongodb://localhost:27017"

MongoClient.connect(url)
.then(client=>{
```

```
const db = client.db("collegeDB")
const collection = db.collection("students")

collection.insertOne({
  name:"Rahul",
  age:21
})

console.log("Data Inserted")

})
```